# THE INDUCTION STEP FROM PROPERTY V-N TO PROPERTY V-N-PLUS-ONE IN THE COMPARISON OF TOPOLOGIES

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The counted statement reads 'In Situation ... we have (V_n) implies (V_{n+1})', so the situation environment of the comparison section is delivered as a context span (source lines 6840-6879); it is the author's own text copied verbatim and makes the card self-contained.

*Editorial note.* The proof is the author's complete argument: the four stages above are all present inside the counted span, and the closing reduction is carried through rather than summarised.

*Editorial note.* Every labelled reference occurring inside the counted spans points to a chapter of the Project that is not part of this card's source tree. Each is carried in assembly.external_references as a non-load-bearing navigation pointer whose source location is the invocation inside the counted span, so no load-bearing mathematics is replaced by a stub.

*Editorial note.* The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble, which declares the situation environment used by the context span.

*Allowed standard prerequisites (editorial).*

- The situation of the comparison section: the site, the two topologies tau and tau', the set of arrows P and the weak Serre subcategories A'_X. Same chapter, the situation environment of the comparison section, delivered verbatim as a context span in this card (source lines 6840-6879). Supplies every object and hypothesis occurring in the statement and in the proof.
- Property (V_n) of the comparison. Same chapter, the property (V_n) stating that every tau'-cohomology class of degree n on an object of C dies after restricting to the members of a tau'-covering and pulling back along an arrow of P. Supplies the hypothesis from which (V_{n+1}) is deduced.
- Locality of cohomology and the cohomology comparison in degree n. Same chapter, the locality lemma for cohomology classes, the comparison result identifying tau'-classes on X after pullback along an arrow of P, and the Leray-type lemma used at the end. Supplies the two reductions at the beginning of the proof and the final computation with the distinguished triangle attached to F -> Rf_{tau,*}f_tau^{-1}F.

*Source reference.* Original section 0EWK: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 6177–6177. Navigation pointer invoked at the counted span; the counted span contains the pointer only and the cited result

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*Source reference.* Original lemma 072W: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 747–747. Navigation pointer invoked at proof line 7157; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

*Source reference.* Original lemma 01FW: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 715–715. Navigation pointer invoked at proof line 7163; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

*Source reference.* Original lemma 0EZC: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 7061–7061. Navigation pointer invoked at proof line 7184; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

*Source reference.* Original lemma 0EZD: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 7090–7090. Navigation pointer invoked at proof line 7191; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

*Source reference.* Original lemma 0D6H: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 4403–4403. Navigation pointer invoked at proof line 7209; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

*Source reference.* Original lemma 08J5: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 7212–7212. Navigation pointer invoked at proof line 7212; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

*Source reference.* Original equation 0EZ2: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 6817–6817. Navigation pointer invoked at proof line 7225; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

*Source reference.* Original lemma 0EZA: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 6914–6914. Navigation pointer invoked at proof line 7231; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

*Source reference.* Original lemma 0EZ9: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 6882–6882. Navigation pointer invoked at proof line 7245; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

*Source reference.* Original lemma 0EZB: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 7029–7029. Navigation pointer invoked at proof line 7275; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

**Situation 0.1.** With $\mathcal{C}$, $\tau$, and $\tau'$ as in Section 0EWK. Assume we are given a subset $\mathcal{P} \subset \text{Arrows}(\mathcal{C})$ and for every object $X$ of $\mathcal{C}$ we are given a weak Serre subcategory $\mathcal{A}'_X \subset Ab(\mathcal{C}_{\tau'}/X)$. We make the following assumption:

(1) given $f : X \to Y$ in $\mathcal{P}$ and $Y' \to Y$ general, then $X \times_Y Y'$ exists and $X \times_Y Y' \to Y'$ is in $\mathcal{P}$,
(2) $f_{\tau'}^{-1}$ sends $\mathcal{A}'_Y$ into $\mathcal{A}'_X$ for any morphism $f : X \to Y$ of $\mathcal{C}$,
(3) given $X$ in $\mathcal{C}$ and $\mathcal{F}'$ in $\mathcal{A}'_X$, then $\mathcal{F}'$ satisfies the sheaf condition for $\tau$-coverings, i.e., $\mathcal{F}' = \epsilon_{X,*}\epsilon_X^{-1}\mathcal{F}'$,
(4) if $f : X \to Y$ in $\mathcal{P}$ and $\mathcal{F}' \in \text{Ob}(\mathcal{A}'_X)$, then $R^i f_{\tau',*}\mathcal{F}' \in \text{Ob}(\mathcal{A}'_Y)$ for $i \geq 0$.
(5) if $\{U_i \to U\}_{i\in I}$ is a $\tau$-covering, then there exist
   (a) a $\tau'$-covering $\{V_j \to U\}_{j\in J}$,
   (b) a $\tau$-covering $\{f_j : W_j \to V_j\}$ consisting of a single $f_j \in \mathcal{P}$, and
   (c) a $\tau'$-covering $\{W_{jk} \to W_j\}_{k\in K_j}$
   such that $\{W_{jk} \to U\}_{j\in J, k\in K_j}$ is a refinement of $\{U_i \to U\}_{i\in I}$.

**Lemma 0.2.** *In Situation 0.1 we have* $(V_n) \Rightarrow (V_{n+1})$.

*Proof.* Let $X$ in $\mathcal{C}$ and $\mathcal{F}$ in $\mathcal{A}_X$. Let $\xi \in H^{n+1}_\tau(U, \mathcal{F})$ for some $U/X$. We have to show that $\xi$ restricts to zero on the members of a $\tau'$-covering of $U$. See Lemma 072W. It follows from this that we may replace $U$ by the members of a $\tau'$-covering of $U$.

By locality of cohomology (Lemma 01FW) we can choose a $\tau$-covering $\{U_i \to U\}$ such that $\xi$ restricts to zero on $U_i$. Choose $\{V_j \to V\}$, $\{f_j : W_j \to V_j\}$, and $\{W_{jk} \to W_j\}$ as in (5). After replacing both $U$ by $V_j$ and $\mathcal{F}$ by its restriction to $\mathcal{C}_\tau/V_j$, which is allowed by (1), we reduce to the case discussed in the next paragraph.

Here $f : X \to Y$ is an element of $\mathcal{P}$ such that $\{X \to Y\}$ is a $\tau$-covering, $\mathcal{F}$ is an object of $\mathcal{A}_Y$, and $\xi \in H^{n+1}_\tau(Y, \mathcal{F})$ is such that there exists a $\tau'$-covering $\{X_i \to X\}_{i\in I}$ such that $\xi$ restricts to zero on $X_i$ for all $i \in I$. Problem: show that $\xi$ restricts to zero on a $\tau'$-covering of $Y$.

By Lemma 0EZC there exists a unique $\tau'$-cohomology class $\theta \in H^{n+1}_{\tau'}(X, \epsilon_{X,*}\mathcal{F})$ whose image is $\xi|_X$. Since $\xi|_X$ pulls back to the same class on $X \times_Y X$ via the two projections, we find that the same is true for $\theta$ (by uniqueness). By Lemma 0EZD we see that after replacing $Y$ by the members of a $\tau'$-covering, we may assume that $\theta = 0$. Consequently, we may assume that $\xi|_X$ is zero.

Let $f : X \to Y$ be an element of $\mathcal{P}$ such that $\{X \to Y\}$ is a $\tau$-covering, $\mathcal{F}$ is an object of $\mathcal{A}_Y$, and $\xi \in H^{n+1}_\tau(Y, \mathcal{F})$ maps to zero in $H^{n+1}_\tau(X, \mathcal{F})$. Problem: show that $\xi$ restricts to zero on a $\tau'$-covering of $Y$.

The assumptions tell us $\xi$ maps to zero under the map

$$\mathcal{F} \longrightarrow Rf_{\tau,*}f_\tau^{-1}\mathcal{F}$$

Use Lemma 0D6H. A simple argument using the distinguished triangle of truncations (Derived Categories, Remark 08J5) shows that $\xi$ maps to zero under the map

$$\mathcal{F} \longrightarrow \tau_{\leq n}Rf_{\tau,*}f_\tau^{-1}\mathcal{F}$$

We will compare this with the map $\epsilon_{Y,*}\mathcal{F} \to K$ where

$$K = \tau_{\leq n} Rf_{\tau',*} f_{\tau'}^{-1} \epsilon_{Y,*}\mathcal{F} = \tau_{\leq n} Rf_{\tau',*} \epsilon_{X,*} f_\tau^{-1}\mathcal{F}$$

The equality $\epsilon_{X,*} f_\tau^{-1} = f_{\tau'}^{-1}\epsilon_{Y,*}$ is a property of (0EZ2). Consider the map

$$Rf_{\tau',*}\epsilon_{X,*} f_\tau^{-1}\mathcal{F} \longrightarrow Rf_{\tau',*} R\epsilon_{X,*} f_\tau^{-1}\mathcal{F} = R\epsilon_{Y,*} Rf_{\tau,*} f_\tau^{-1}\mathcal{F}$$

used in the proof of Lemma 0EZA which induces by adjunction a map

$$\epsilon_Y^{-1} Rf_{\tau',*}\epsilon_{X,*} f_\tau^{-1}\mathcal{F} \to Rf_{\tau,*} f_\tau^{-1}\mathcal{F}$$

Taking trunctions we find a map

$$\epsilon_Y^{-1} K \longrightarrow \tau_{\leq n} Rf_{\tau,*} f_\tau^{-1}\mathcal{F}$$

which is an isomorphism by Lemma 0EZA; the lemma applies because $f_\tau^{-1}\mathcal{F}$ is in $\mathcal{A}_X$ by Lemma 0EZ9. Choose a distinguished triangle

$$\epsilon_{Y,*}\mathcal{F} \to K \to L \to \epsilon_{Y,*}\mathcal{F}[1]$$

The map $\mathcal{F} \to f_{\tau,*} f_\tau^{-1}\mathcal{F}$ is injective as $\{X \to Y\}$ is a $\tau$-covering. Thus $\epsilon_{Y,*}\mathcal{F} \to \epsilon_{Y,*} f_{\tau,*} f_\tau^{-1}\mathcal{F} = f_{\tau',*} f_{\tau'}^{-1}\epsilon_{Y,*}\mathcal{F}$ is injective too. Hence $L$ only has nonzero cohomology sheaves in degrees $0, \ldots, n$. As $f_{\tau',*} f_{\tau'}^{-1}\epsilon_{Y,*}\mathcal{F}$ is in $\mathcal{A}'_Y$ by (2) and (4) we conclude that

$$H^0(L) = \mathrm{Coker}(\epsilon_{Y,*}\mathcal{F} \to f_{\tau',*} f_{\tau'}^{-1}\epsilon_{Y,*}\mathcal{F})$$

is in the weak Serre subcategory $\mathcal{A}'_Y$. For $1 \leq i \leq n$ we see that $H^i(L) = R^i f_{\tau',*} f_{\tau'}^{-1}\epsilon_{Y,*}\mathcal{F}$ is in $\mathcal{A}'_Y$ by (2) and (4). Pulling back the distinguished triangle above by $\epsilon_Y$ we get the distinguished triangle

$$\mathcal{F} \to \tau_{\leq n} Rf_{\tau,*} f_\tau^{-1}\mathcal{F} \to \epsilon_Y^{-1} L \to \mathcal{F}[1]$$

Since $\xi$ maps to zero in the middle term we find that $\xi$ is the image of an element $\xi' \in H^n_\tau(Y, \epsilon_Y^{-1} L)$. By Lemma 0EZB we have

$$H^n_{\tau'}(Y, L) = H^n_\tau(Y, \epsilon_Y^{-1} L),$$

Thus we may lift $\xi'$ to an element of $H^n_{\tau'}(Y, L)$ and take the boundary into $H^{n+1}_{\tau'}(Y, \epsilon_{Y,*}\mathcal{F})$ to see that $\xi$ is in the image of the canonical map $H^{n+1}_{\tau'}(Y, \epsilon_{Y,*}\mathcal{F}) \to H^{n+1}_\tau(Y, \mathcal{F})$. By locality of cohomology for $H^{n+1}_{\tau'}(Y, \epsilon_{Y,*}\mathcal{F})$, see Lemma 01FW, we conclude. $\square$